

Executive Summary — Surrogate Model Validation Report

Use case: UCFATIGUE_1 | 28 July 2026 | 2 model(s) evaluated

REQUIREMENTS PARTIALLY MET

Recommended model: GradientBoosting

Avg R²: 0.9643 — Excellent

Q90 accuracy: 5/7 outputs — 5/7 outputs pass — partially met

Overfitting check: 2/7 outputs — 2/7 outputs pass — overfitting present

Project Overview

Use case	UCFATIGUE_1
Inputs	FLAP, Altitude, TAS, Mass, q, gamma, Type_segment, Xcg(%CMA)
Outputs	7: 1g, Vertical maneuver, Vertical gust, Turn, Frontal gust, n0, Giro
Train / Val / Test	80% / 10% / 10%
Accuracy target	Q90 < 10% relative error per output

Model Performance — GradientBoosting (Test Set)

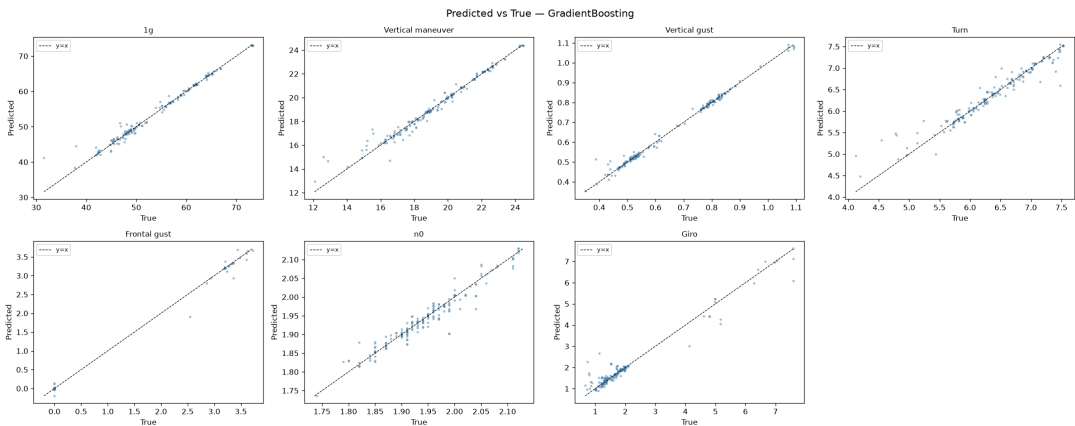
Output	R ²	Q90	Target	Gap	Status
1g	0.9797	0.0257	10%	-0.0743	PASS
Vertical maneuver	0.9723	0.0318	10%	-0.0682	PASS
Vertical gust	0.9914	0.0262	10%	-0.0738	PASS
Turn	0.9286	0.0351	10%	-0.0649	PASS
Frontal gust	0.9964	0.2795	10%	+0.1795	FAIL
n0	0.9317	0.0149	10%	-0.0851	PASS
Giro	0.9499	0.2215	10%	+0.1215	FAIL

Q90 < 10% Q90 ≥ 10% R² ≥ 0.95 R² ∈ [0.80, 0.95) Gap = Q90 – target (negative ⇒ margin)

Analysis & Recommendations

- Significant failures (Frontal gust, Giro): Q90 exceeds target by ≥ 5%. Investigate input features and model capacity.
- Overfitting detected (1g, Vertical gust, Turn, n0, Giro): KS test rejects equal residual distributions on train and test.

Predicted vs True — GradientBoosting (Test Set, sample of 5 000 points)



Overfitting Check — KS Test p-values (GradientBoosting)

H_0 : residuals on train and test follow the same distribution. $p \geq 0.05$ = no overfitting. $p < 0.05$ = overfitting.

Output	KS p-value
lg	0.004
Vertical maneuver	0.161
Vertical gust	0.042
Turn	0.044
Frontal gust	0.974
n0	0.032
Giro	0.017

Full technical details in the [Technical Appendix](#) (page 4).

Technical Appendix

UCFATIGUE_1 — Full Validation Report

Technical Appendix — Full Validation Report — UCFATIGUE_1

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Introduction

This appendix documents the full validation of surrogate models for use case UCFATIGUE_1. A total of 2 model architecture(s) were trained and evaluated: MLP, GradientBoosting. The models predict 7 output quantities (1g, Vertical maneuver, Vertical gust, Turn, Frontal gust, n0, Giro) from 8 input features (FLAP, Altitude, TAS, Mass, q, gamma, Type_segment, Xcg(%CMA)). The accuracy criterion requires that the 90th-percentile relative error (Q90) remains below 10% for every output.

Model Selection

GradientBoosting achieved the highest average R^2 on the test set (0.9643), outperforming MLP ($R^2=0.5237$). It is therefore the recommended model for production deployment.

Model	Avg R^2	Q90 pass (7 outputs)	KS pass (7 outputs)
MLP	0.5237	5/7	7/7
GradientBoosting ★	0.9643	5/7	2/7

★ Recommended model. Q90 target: < 10% relative error at 90th percentile. KS: $p \geq 0.05$ = no overfitting detected.

Accuracy Assessment (Q90 & R^2)

The recommended model GradientBoosting satisfies Q90 on 5 out of 7 outputs. Passing outputs: 1g, Vertical maneuver, Vertical gust, Turn, n0. Failing outputs: **Frontal gust**, **Giro**. See the Analysis & Recommendations section in the Executive Summary for targeted improvement actions.

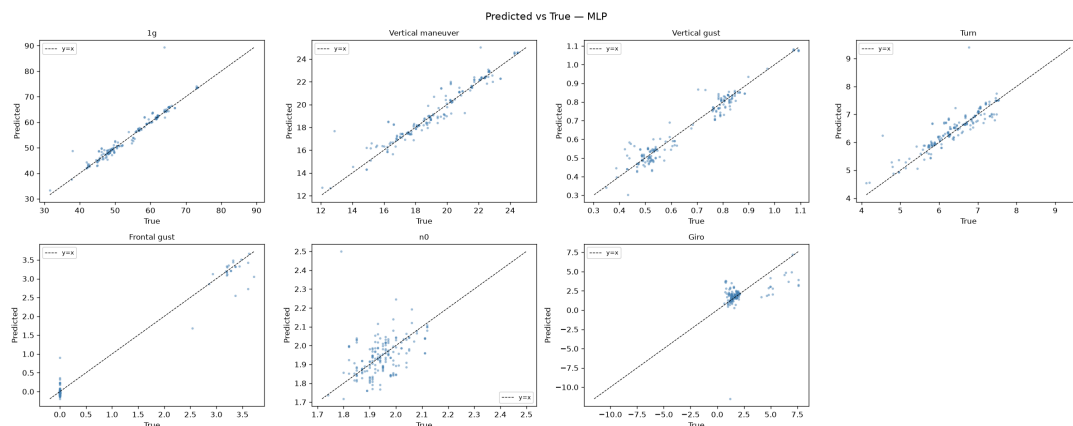
Output	MLP			GradientBoosting		
	R^2	Q90	Gap	R^2	Q90	Gap
1g	0.9160	0.0473	-0.0527	0.9797	0.0257	-0.0743
Vertical maneuver	0.9116	0.0554	-0.0446	0.9723	0.0318	-0.0682
Vertical gust	0.9377	0.0918	-0.0082	0.9914	0.0262	-0.0738
Turn	0.7544	0.0783	-0.0217	0.9286	0.0351	-0.0649
Frontal gust	0.9794	11.3300	+11.2300	0.9964	0.2795	+0.1795
n0	-0.7292	0.0723	-0.0277	0.9317	0.0149	-0.0851
Giro	-0.1042	0.9005	+0.8005	0.9499	0.2215	+0.1215

■ $R^2 \geq 0.95$ ■ $R^2 \in [0.80, 0.95)$ ■ $R^2 < 0.80$ or Q90 fail Gap = Q90 - target (negative $\Rightarrow$ margin above target)

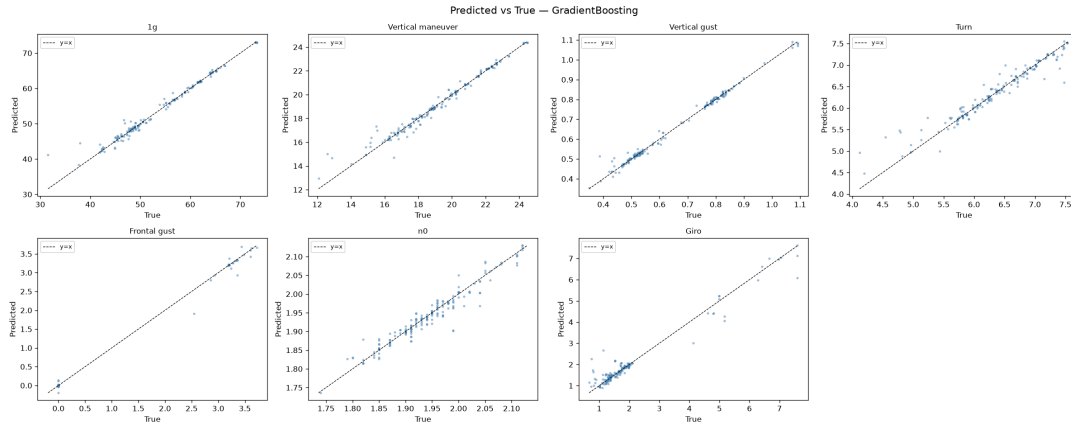
Predicted vs True (Scatter Plots, Test Set)

Each scatter plot displays a random sample of up to 5 000 test-set points. The dashed diagonal represents perfect prediction ($\hat{y} = y$). Systematic bias (points consistently above or below the diagonal) or fan-shaped spread (heteroscedasticity) are indicators of model limitations.

MLP



GradientBoosting



Overfitting Assessment (KS Test)

The Kolmogorov–Smirnov (KS) test compares the empirical distribution of training residuals with that of test residuals. H_0 : both distributions are identical (no overfitting). A p-value below 0.05 provides statistical evidence of overfitting.

The KS test detects distributional differences between train and test residuals for GradientBoosting on the following outputs: **1g**, **Vertical gust**, **Turn**, **n0**, **Giro**. This suggests localised overfitting. Increasing regularisation or reducing model capacity for these outputs is recommended before deployment.

Output	MLP	GradientBoosting
1g	0.473	0.004
Vertical maneuver	0.180	0.161
Vertical gust	0.635	0.042
Turn	0.078	0.044
Frontal gust	0.736	0.974
n0	0.378	0.032
Giro	0.781	0.017

■ $p \geq 0.05$ — residual distributions consistent
 ■ $p < 0.05$ — overfitting detected

Data Split Quality (VTP Analysis)

The Voronoi Tessellation Proximity (VTP) method evaluates whether the test set is representative of, and well-covered by, the training distribution. A low residual voxel proportion indicates that test points are not isolated outside the training space.

The Voronoi Tessellation Proximity (VTP) analysis confirms that the test set is well-distributed within the training space (residual voxel proportion ≤ 0.05 , no isolated test points). The chi-squared test ($p \geq 0.05$) validates that the split is statistically unbiased. The train/test partition is of high quality.

Metric	Value	Pass
Residual voxel proportion (≤ 0.05)	0.000	✓
Valid test proportion	0.931	
Phacking test proportion	0.006	
Isolated test proportion	0.063	
Chi ² p-value (≥ 0.05)	0.7969	✓

Improvement Roadmap

The following actions are prioritised based on the failure analysis above.

- Significant failures (Frontal gust, Giro): Q90 exceeds target by $\geq 5\%$. Investigate input features and model capacity. Consider whether additional physical drivers of these outputs are missing from the input feature set,

or evaluate a more expressive architecture (e.g. deeper network, ensemble).

- Overfitting detected (1g, Vertical gust, Turn, n0, Giro): KS test rejects equal residual distributions on train and test. Increase regularisation strength (dropout rate, weight decay, early-stopping patience) or reduce model capacity.